ANY Y-IMMERSION OF AN ETALE CHART OF THE SOURCE OF AN LCI MAP OF SPACES INTO A SMOOTH Y-SPACE IS KOSZUL-REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11171-11187 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 11189-11201, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; the A (schemes) counterpart lives in another chapter file (pool A lemma at lines 18589ff is a different label).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source definitions regular-immersion and lci; More on Morphisms, lemma lemma-lci (the schemes A-pool analogue is a separate counted card).

Source reference. Original lemma 069F: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11153–11164. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

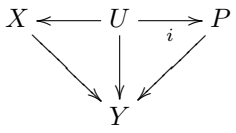
Source reference. Original lemma 06BN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10530–10546. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069G: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11201–11201. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a local complete intersection morphism of algebraic spaces over S. Let P be an algebraic space smooth over Y. Let $U \to X$ be an étale morphism of algebraic spaces and let $i : U \to P$ an*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

immersion of algebraic spaces over Y. Picture:



Then i is a Koszul-regular immersion of algebraic spaces.

Proof. Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme W and a surjective étale morphism $W \to P \times_Y V$. Set $U' = U \times_P W$, which is a scheme étale over U. We have to show that $U' \to W$ is a Koszul-regular immersion of schemes, see Definition 06BN. By Definition 069F above the morphism of schemes $U' \to V$ is a local complete intersection morphism. Hence the result follows from More on Morphisms, Lemma 069G. □